

Embeddings and retrieval augmented generation

Foundations of AI

Carlos Cotrini

UZH CAS Gamechanger AI

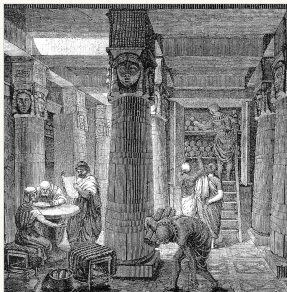
26 September 2026

1

Section 1

Indexes before computers

Alexandria, about 250 BCE



- A scroll has no title page. To learn what it is, you unroll it.

Imagined by O. Von Corven, 19th century.

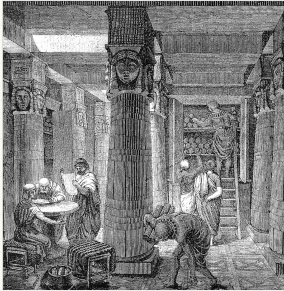
About 250 BCE

121 CE

1230

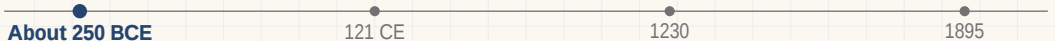
1895

Alexandria, about 250 BCE

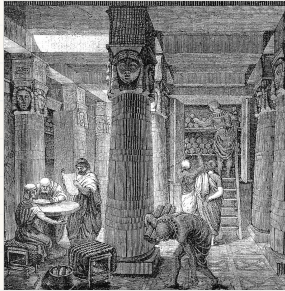


- A scroll has no title page. To learn what it is, you unroll it.
- **A scholar arrives wanting the works of one author.**

Imagined by O. Von Corven, 19th century.

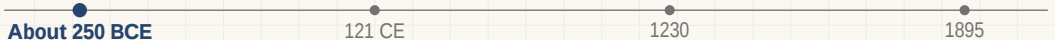


Alexandria, about 250 BCE

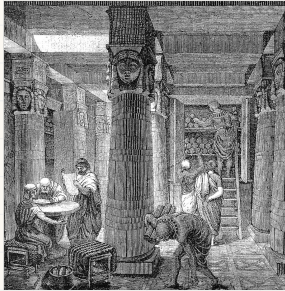


- A scroll has no title page. To learn what it is, you unroll it.
- **A scholar arrives wanting the works of one author.**
- Callimachus wrote **one entry per work: author, opening words, length in lines**, in 120 scrolls.

Imagined by O. Von Corven, 19th century.



Alexandria, about 250 BCE



Imagined by O. Von Corven, 19th century.

- A scroll has no title page. To learn what it is, you unroll it.
- **A scholar arrives wanting the works of one author.**
- Callimachus wrote **one entry per work: author, opening words, length in lines**, in 120 scrolls.
- You read the entries until you know which of **the scrolls** to unroll.

About 250 BCE

121 CE

1230

1895

China, 121 CE



- A writing system with no alphabet has no order to sort by.

A page of the Kangxi Dictionary, 1716.

About 250 BCE

121 CE

1230

1895

China, 121 CE



- A writing system with no alphabet has no order to sort by.
- You meet a character you cannot read.

A page of the Kangxi Dictionary, 1716.

About 250 BCE

121 CE

1230

1895

China, 121 CE



- A writing system with no alphabet has no order to sort by.
- **You meet a character you cannot read.**
- Xu Shen gave each one an address: **one of 540 components**, and after 1615 **a stroke count too**.

A page of the Kangxi Dictionary, 1716.

About 250 BCE

121 CE

1230

1895

China, 121 CE



- A writing system with no alphabet has no order to sort by.
- **You meet a character you cannot read.**
- Xu Shen gave each one an address: **one of 540 components**, and after 1615 **a stroke count too**.
- Follow the address and you reach **the entry: what it means, how it is said.**

A page of the Kangxi Dictionary, 1716.

About 250 BCE

121 CE

1230

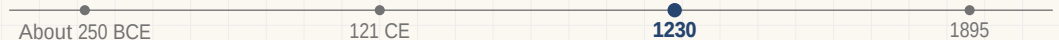
1895

Paris, 1230



- About 500 friars listed every word of the Bible, and everywhere it appears.

Hugh of Saint-Cher. Treviso, 1352.

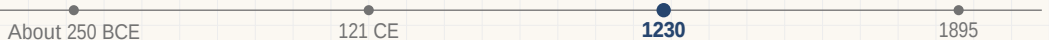


Paris, 1230



- About 500 friars listed every word of the Bible, and everywhere it appears.
- **You want every passage that uses one word.**

Hugh of Saint-Cher. Treviso, 1352.

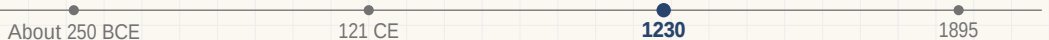


Paris, 1230



- About 500 friars listed every word of the Bible, and everywhere it appears.
- **You want every passage that uses one word.**
- There was no way to say where, so they cut every chapter into **seven parts, marked A to G.**

Hugh of Saint-Cher. Treviso, 1352.



Paris, 1230



Hugh of Saint-Cher. Treviso, 1352.

- About 500 friars listed every word of the Bible, and everywhere it appears.
- **You want every passage that uses one word.**
- There was no way to say where, so they cut every chapter into **seven parts, marked A to G.**
- The address sends you to **the passages themselves.** We still cite that way.

About 250 BCE

121 CE

1230

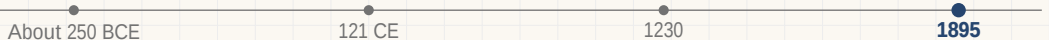
1895

Brussels, 1895



- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.

The Repertoire, about 1900.



Brussels, 1895



The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.
- **From 1896 you could post them a question, for a fee.**

About 250 BCE

121 CE

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1895

Brussels, 1895



The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.
- **From 1896 you could post them a question, for a fee.**
- They searched the cards and mailed back the ones that matched.

About 250 BCE

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Brussels, 1895



The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.
- **From 1896 you could post them a question, for a fee.**
- They searched the cards and mailed back the ones that matched.
- The cards told you where to find **the books and the articles**. Over 1,500 questions a year by 1912.

About 250 BCE

121 CE

1230

1895

The same three things, four times

Query

Keys

Values

The same three things, four times

	Query	Keys	Values
Alexandria, 250 BCE	The works of one author	Author, opening words, length in lines	The scrolls

The same three things, four times

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The same three things, four times

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Brussels, 1895	A question sent by post	Fifteen million index cards	The books and articles
A company, 2026	A question in plain words	?	The company's own documents

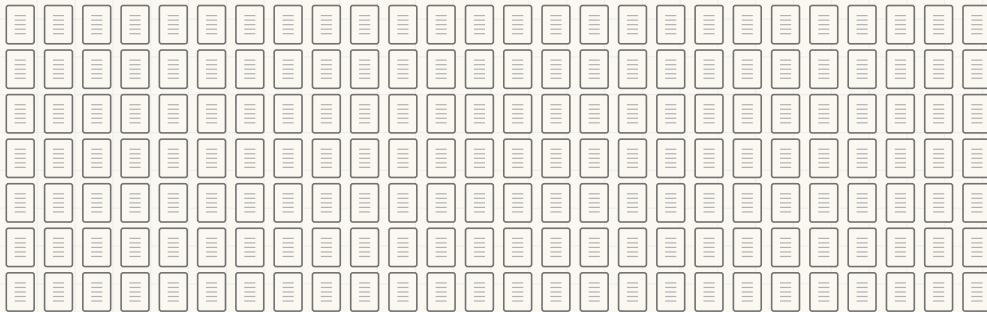
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Section 2

Information retrieval

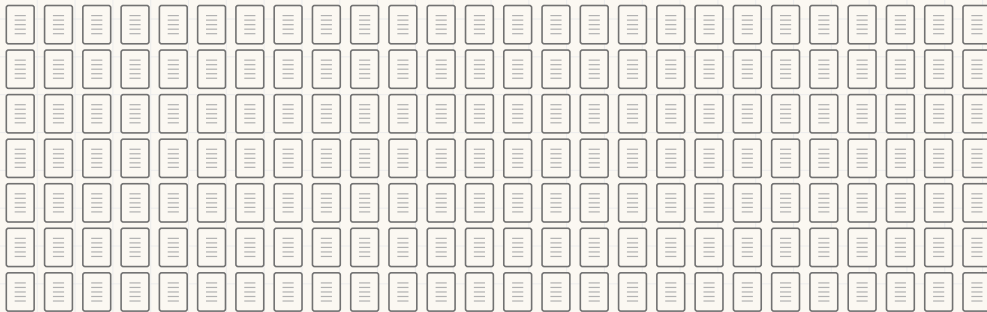
Finding the few documents that answer a question

One company's own documents



Finding the few documents that answer a question

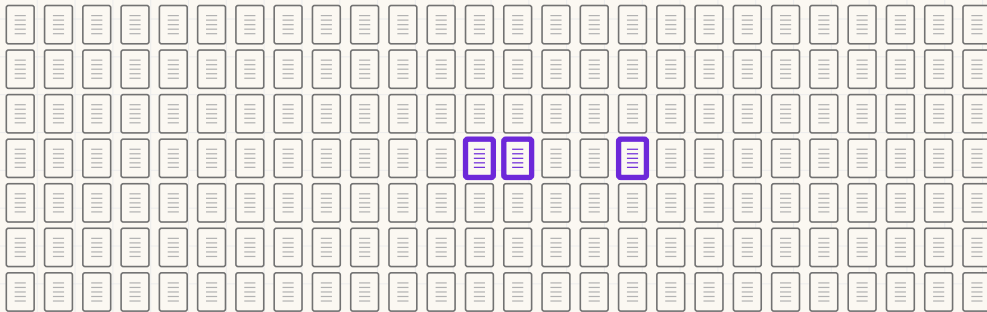
One company's own documents



“What did we agree with the rope supplier about late delivery?”

Finding the few documents that answer a question

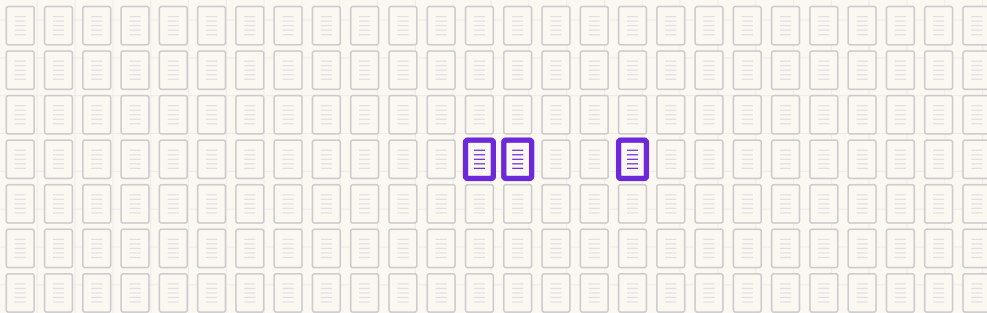
One company's own documents



“What did we agree with the rope supplier about late delivery?”

Finding the few documents that answer a question

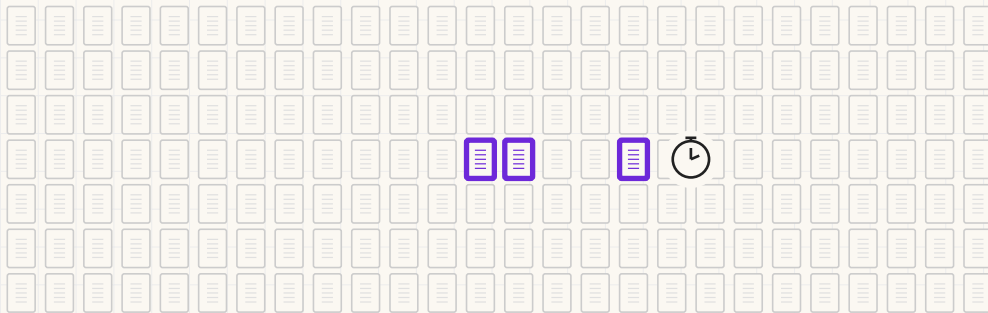
One company's own documents



“What did we agree with the rope supplier about late delivery?”

Finding the few documents that answer a question

One company's own documents



“What did we agree with the rope supplier about late delivery?”

Queries, keys and values

Queries

How much do we get
back for each extra
day a consignment is
overdue?

Keys

Four of the ten keys shown

Incident report
Weather and closure rule
Supplier contract clause
Restaurant notice

Values

The company's own documents



Queries, keys and values

Queries

How much do we get
back for each extra
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Keys

Four of the ten keys shown

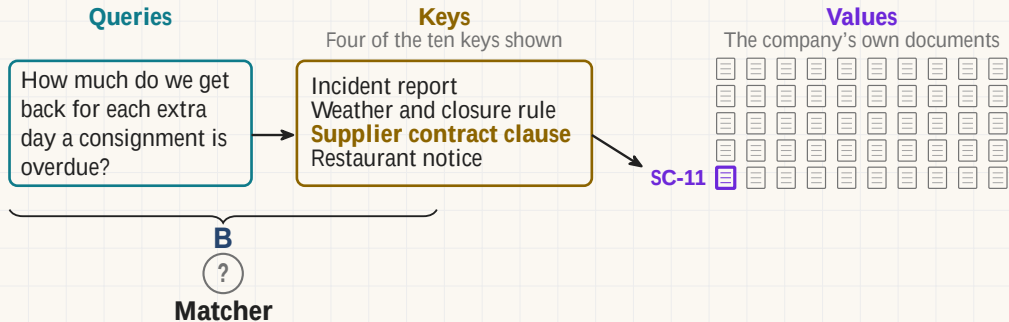
Incident report
Weather and closure rule
Supplier contract clause
Restaurant notice

Values

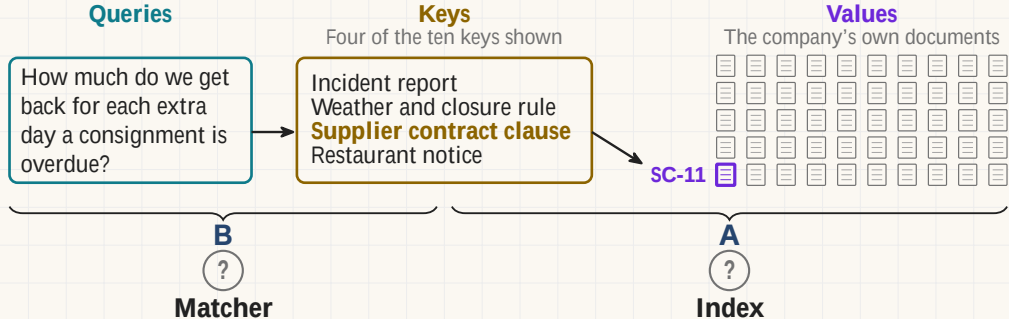
The company's own documents



Queries, keys and values



Queries, keys and values



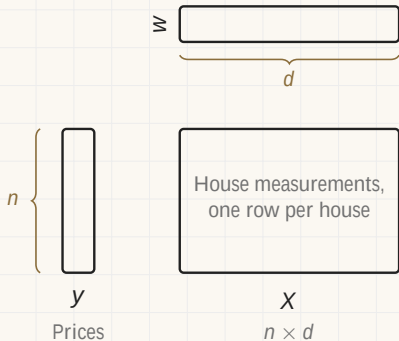
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Section 3

Machine learning needs vectors

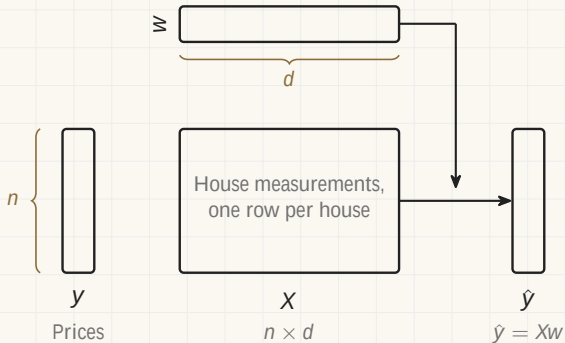
Linear regression with gradient descent

At one o'clock, in one picture. Each box is a piece of data, and its shape is the shape of the data.



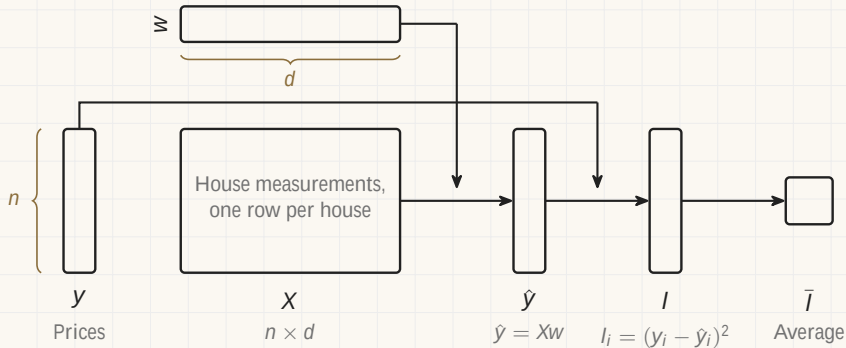
Linear regression with gradient descent

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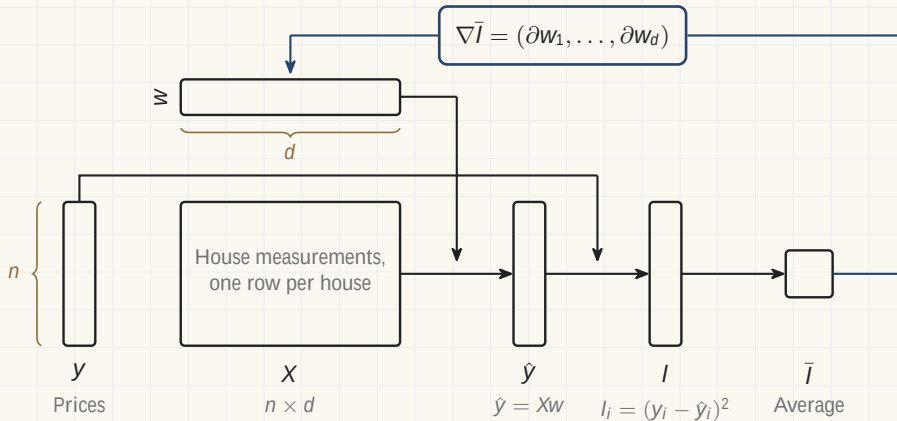
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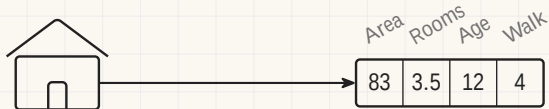


Linear regression with gradient descent

At one o'clock, in one picture. Each box is a piece of data, and its shape is the shape of the data.

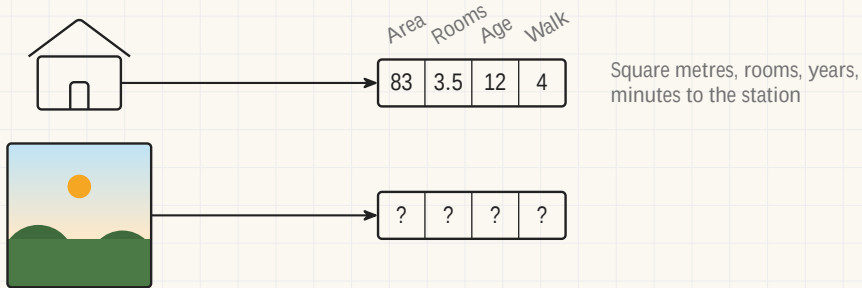


How do we do machine learning with images and text?

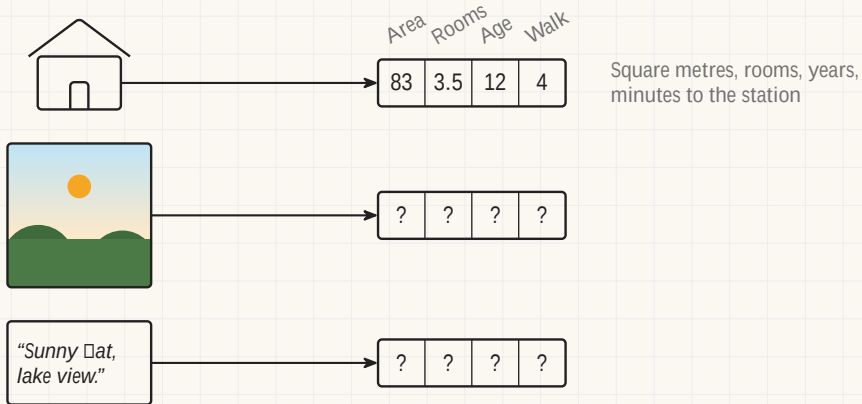


Square metres, rooms, years,
minutes to the station

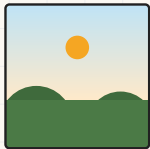
How do we do machine learning with images and text?



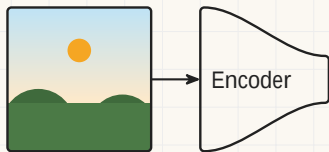
How do we do machine learning with images and text?



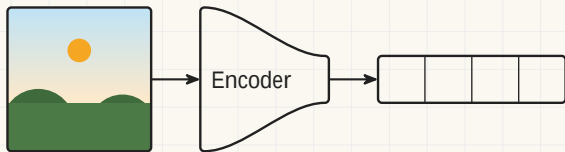
Autoencoders



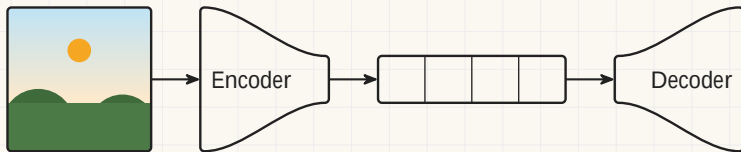
Autoencoders



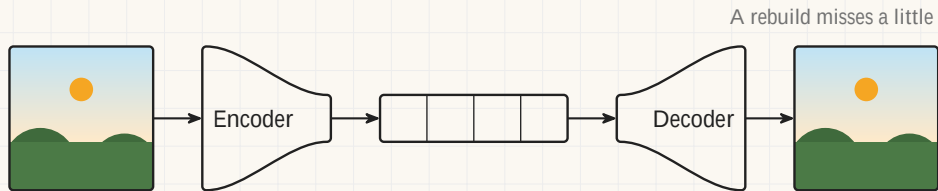
Autoencoders



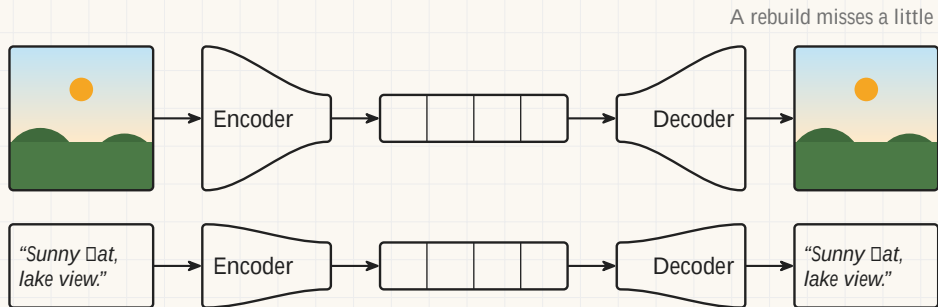
Autoencoders



Autoencoders



Autoencoders

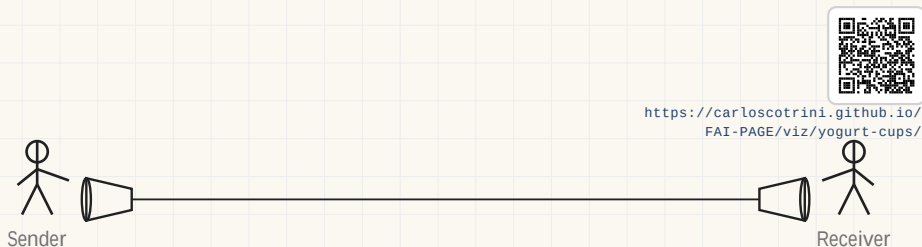


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Section 4

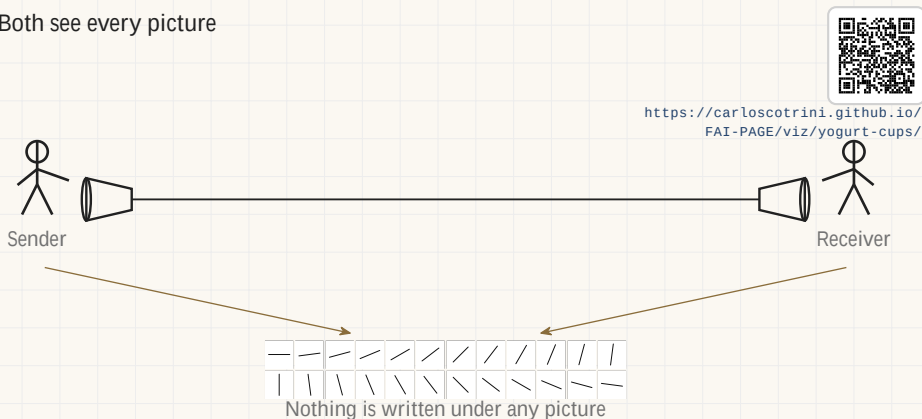
The yogurt cup game

The yogurt cup game



The yogurt cup game

- Both see every picture

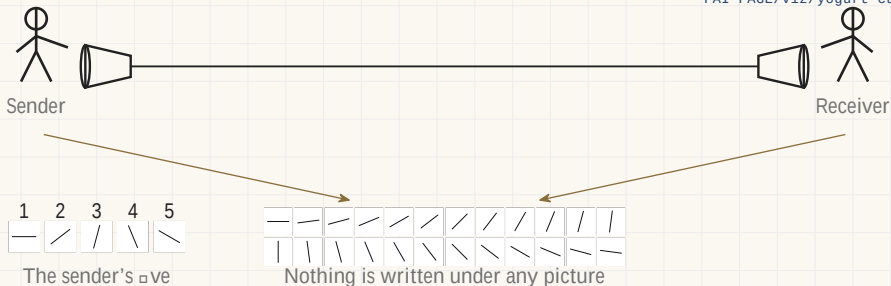


The yogurt cup game

- Both see every picture
- One sees a ve of them, in order



<https://carlostrini.github.io/FAI-PAGE/viz/yogurt-cups/>

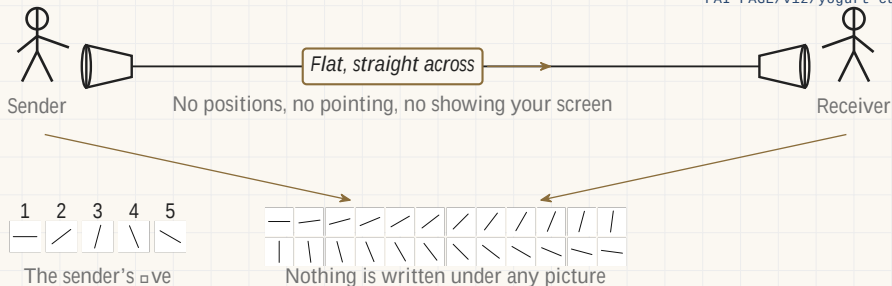


The yogurt cup game

- Both see every picture
- One sees a few of them, in order
- Only words go through the cup



<https://carlostrini.github.io/FAI-PAGE/viz/yogurt-cups/>

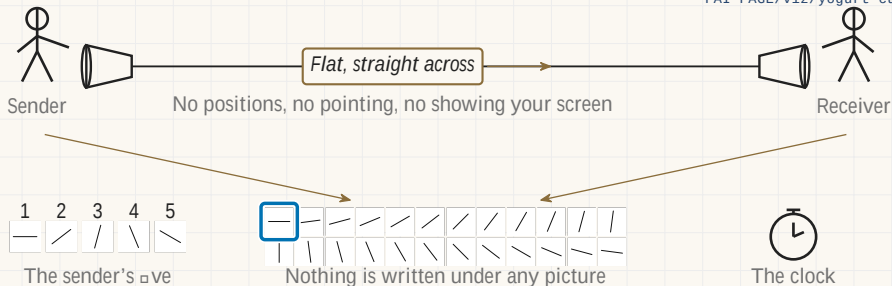


The yogurt cup game

- Both see every picture
- One sees a few of them, in order
- Only words go through the cup
- The partner finds them, against the clock



<https://carloscotrini.github.io/FAI-PAGE/viz/yogurt-cups/>



The score is the time

Round 1

Steep, nearly upright



The clock

The score is the time

Round 1

Steep, nearly upright

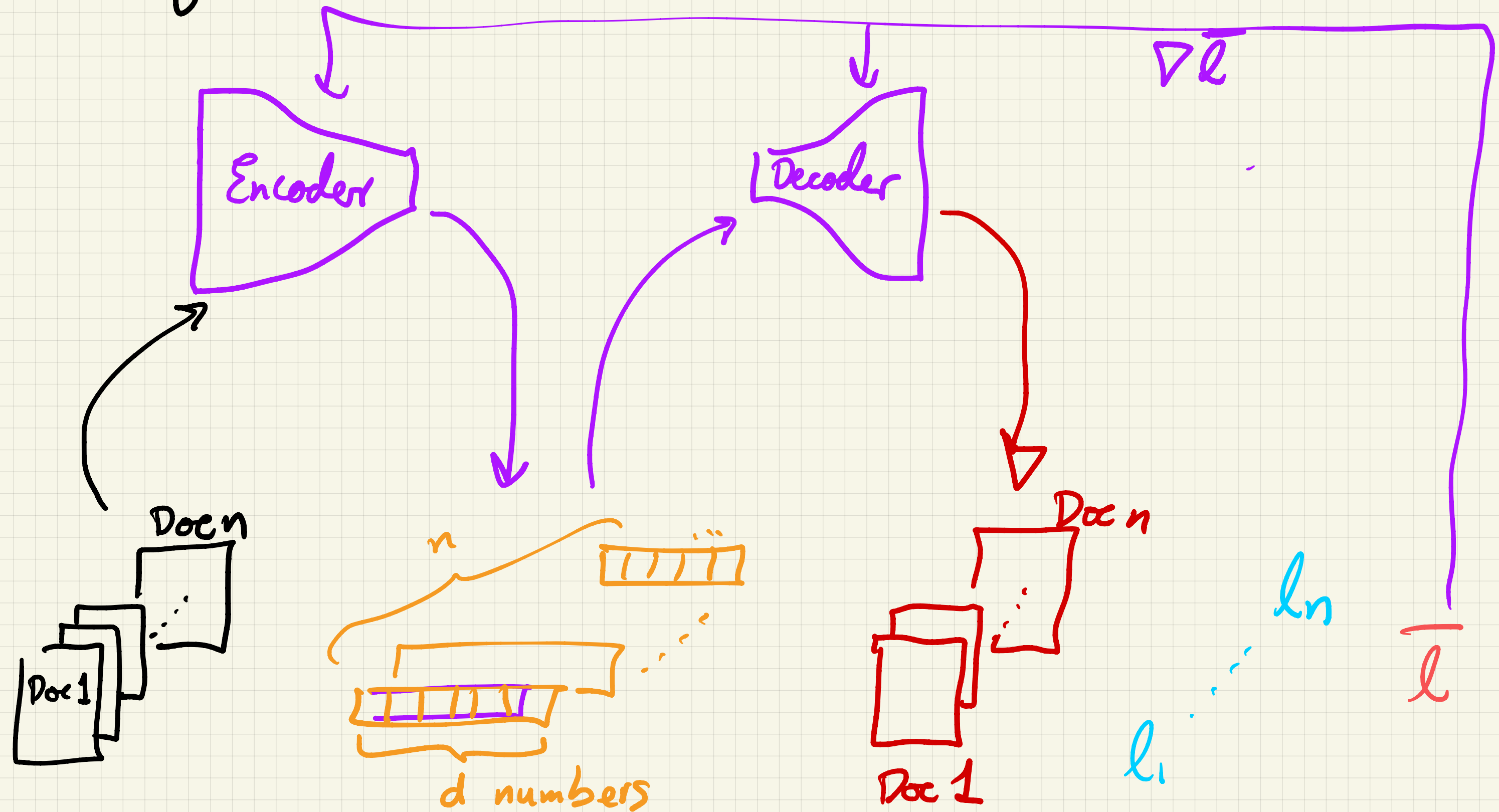


"Not that one. The clock is still running."

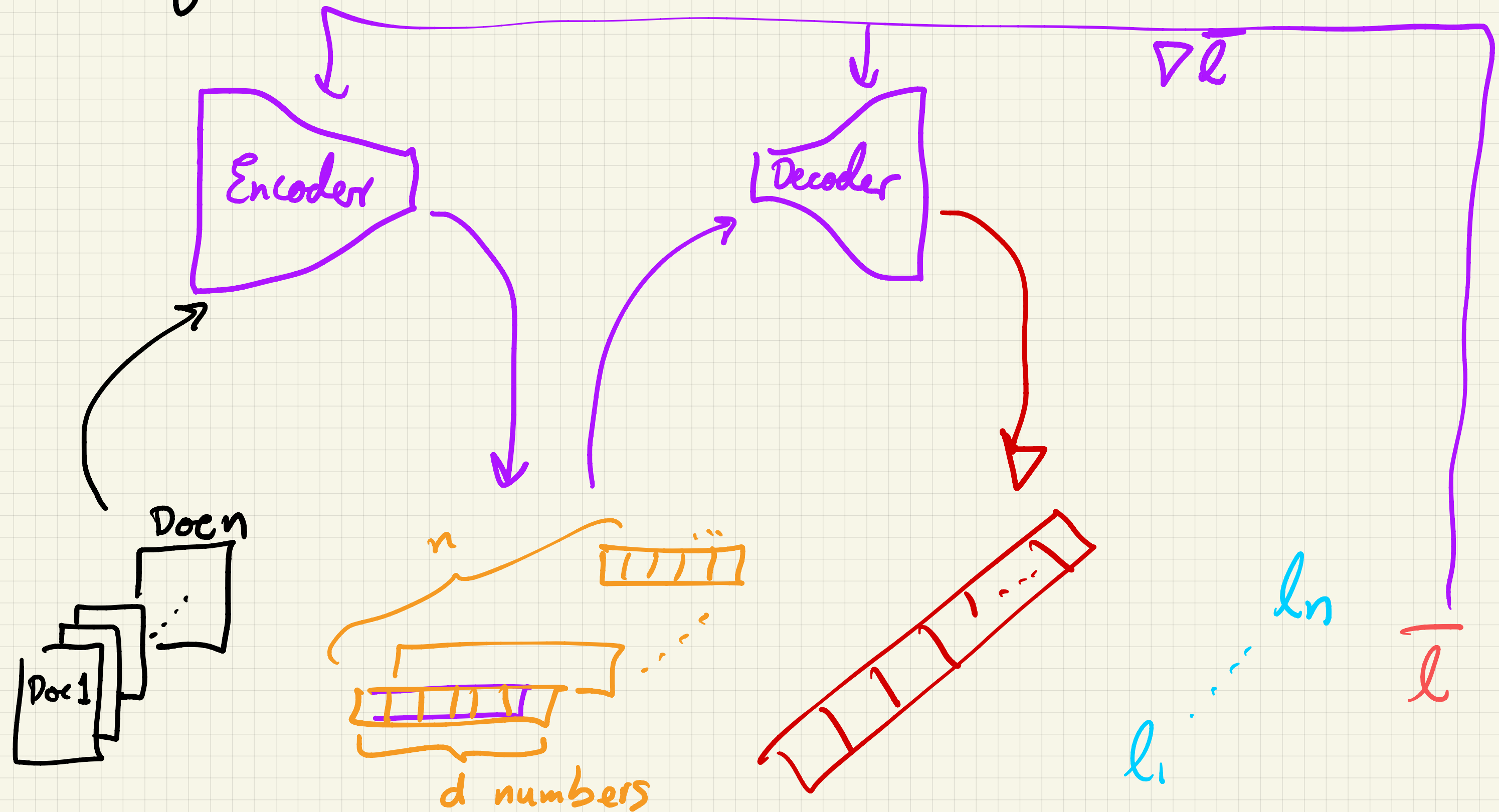


The clock

Building an indexation system with NNs



Building an indexation system with NNs

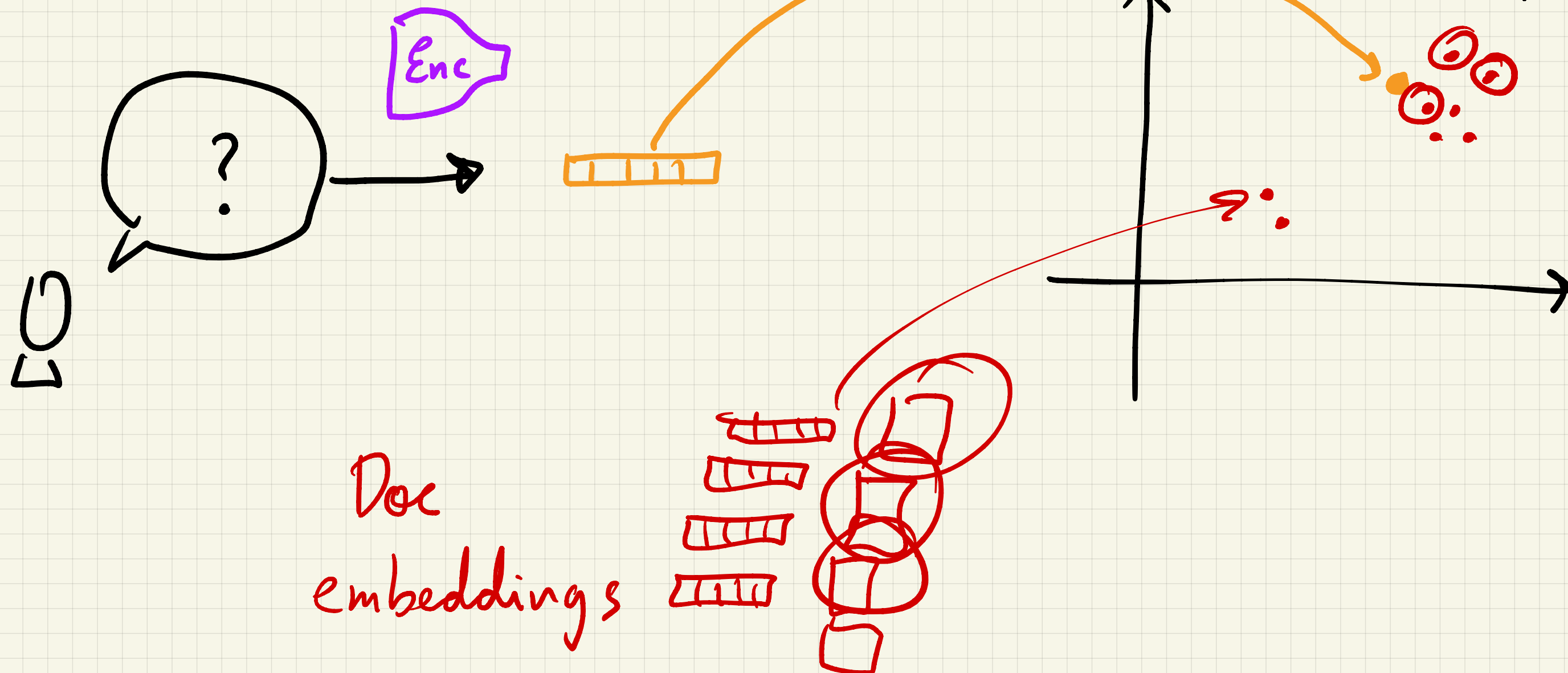


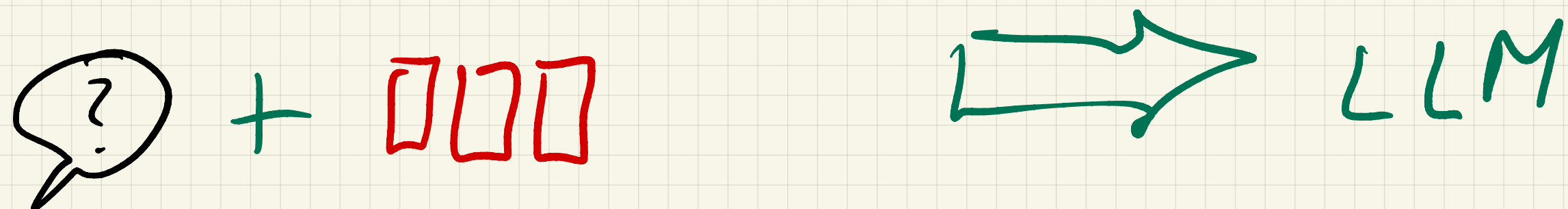
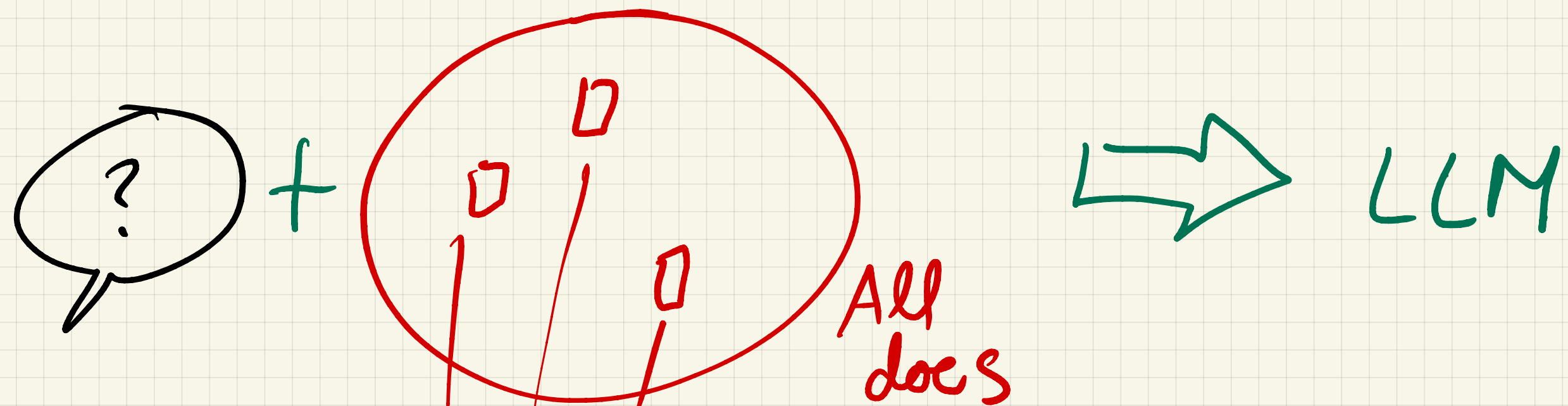
Indices

RAG

to answer it

Representation
Space





The score is the time

Round 1

Steep, nearly upright



"Not that one. The clock is still running."



The clock

The score is the time

Round 1

Steep, nearly upright



"Not that one. The clock is still running."



The clock

Between rounds

*"The pairs who get fast are the ones
who stop describing and start counting."*

The score is the time

Round 1

Steep, nearly upright



"Not that one. The clock is still running."



The clock

Between rounds

"The pairs who get fast are the ones who stop describing and start counting."

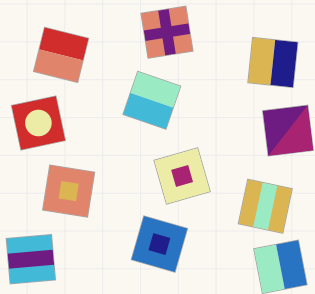
Round 2

Ten steps up from flat



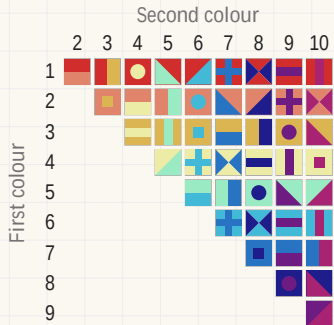
Right, first time

Words are a coarse code



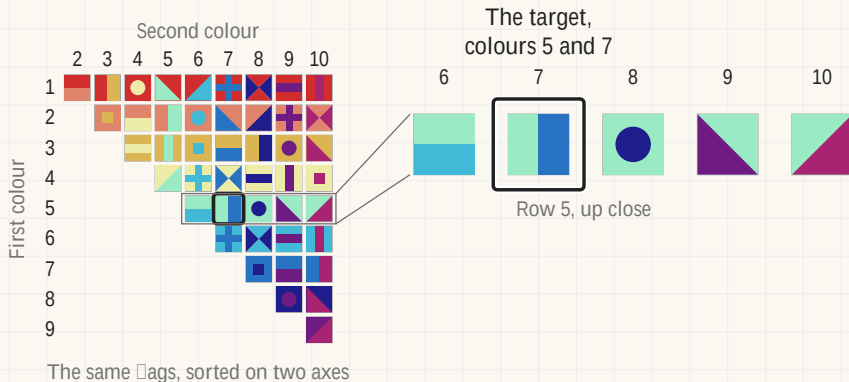
Twelve of them, in no order

Words are a coarse code

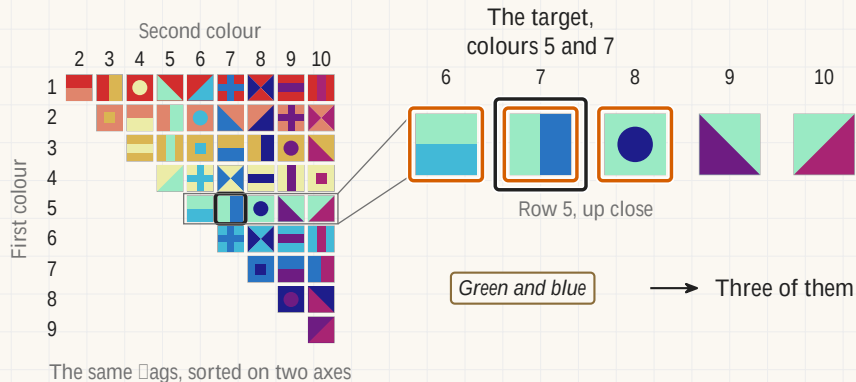


The same flags, sorted on two axes

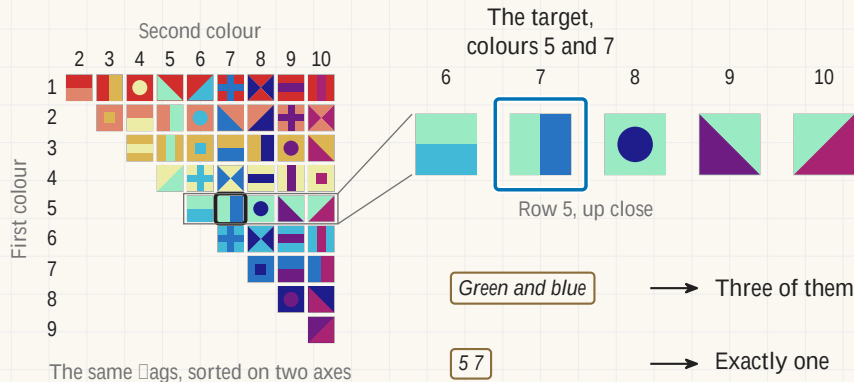
Words are a coarse code



Words are a coarse code

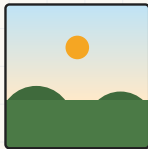


Words are a coarse code



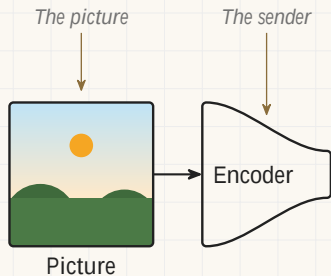
The game, named

The picture

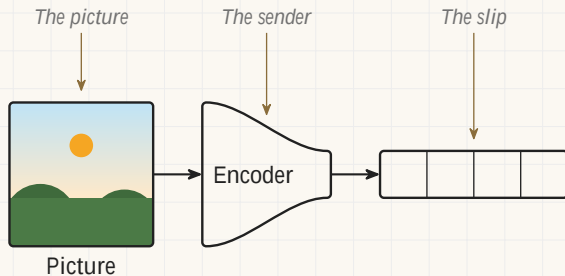


Picture

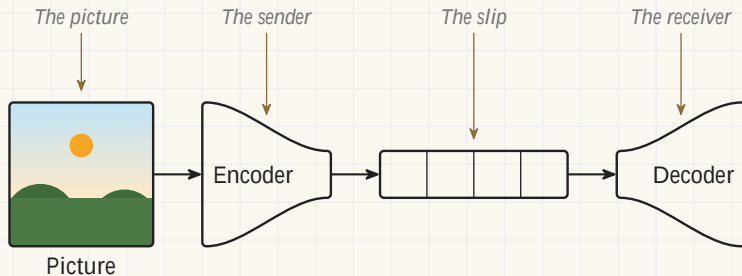
The game, named



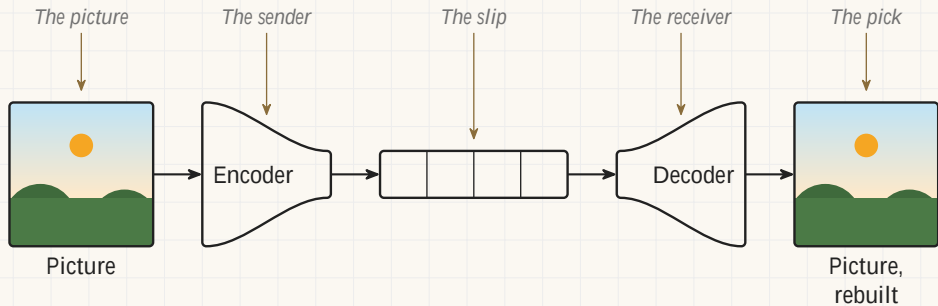
The game, named



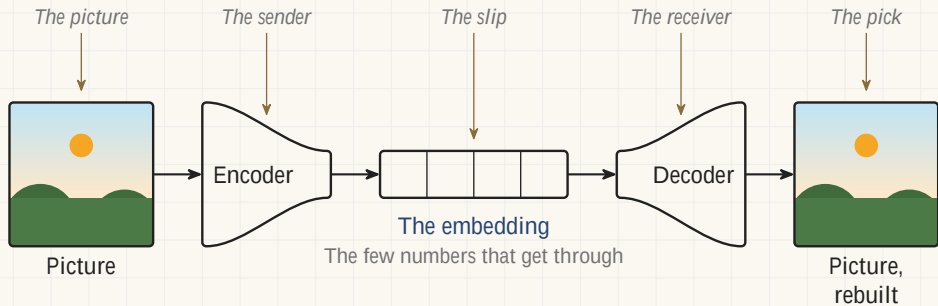
The game, named



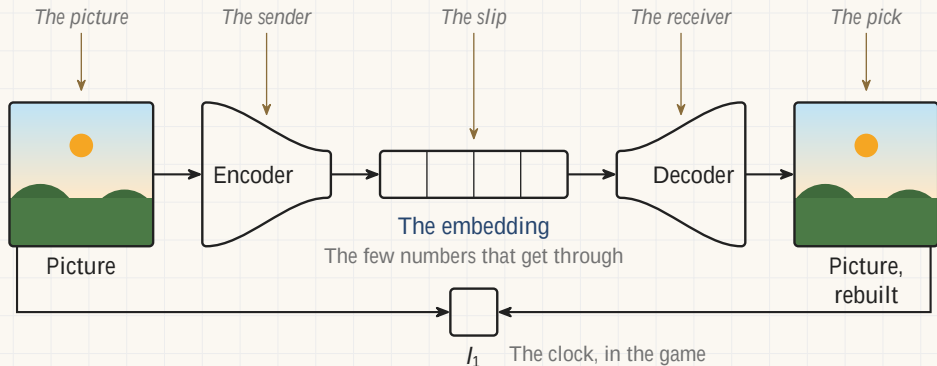
The game, named



The game, named



The game, named



5

Section 5

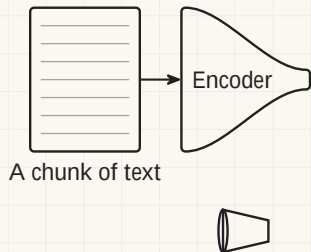
Embeddings of text

An autoencoder for text

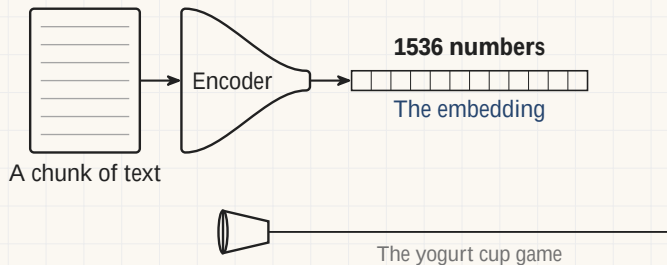


A chunk of text

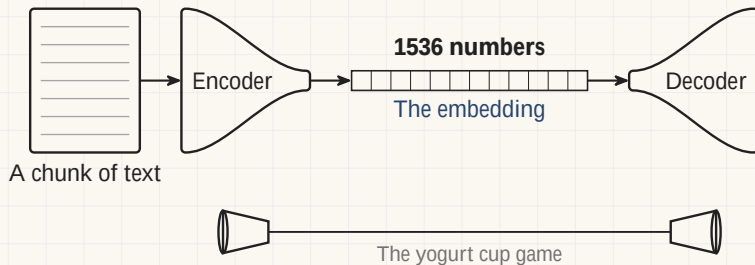
An autoencoder for text



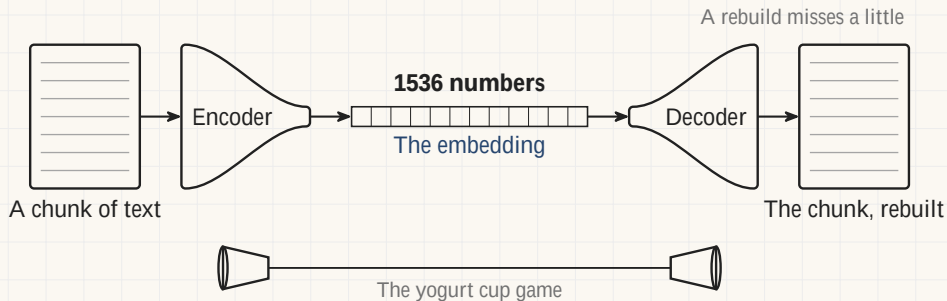
An autoencoder for text



An autoencoder for text



An autoencoder for text



How it is trained

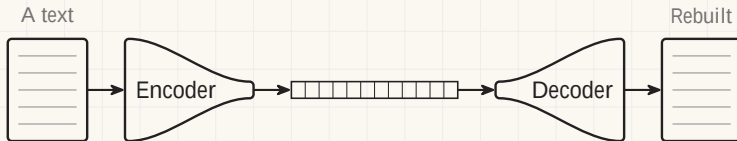
- 1 Collect a large collection of texts

A text



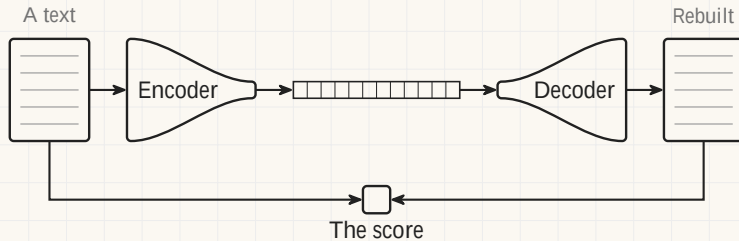
How it is trained

- 1 Collect a large collection of texts
- 2 Initialise the weights at random



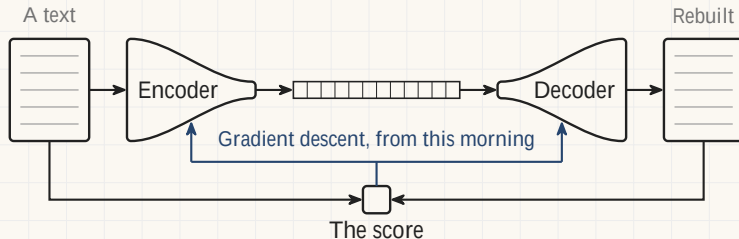
How it is trained

- 1 Collect a large collection of texts
- 2 Initialise the weights at random
- 3 Play the game, take the score



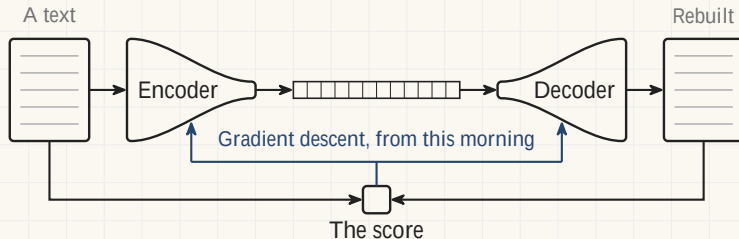
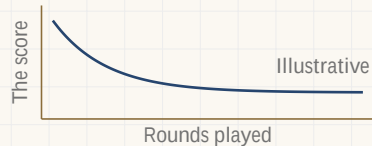
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- 1 Collect a large collection of texts
- 2 Initialise the weights at random
- 3 Play the game, take the score
- 4 Update the weights from the score, play again



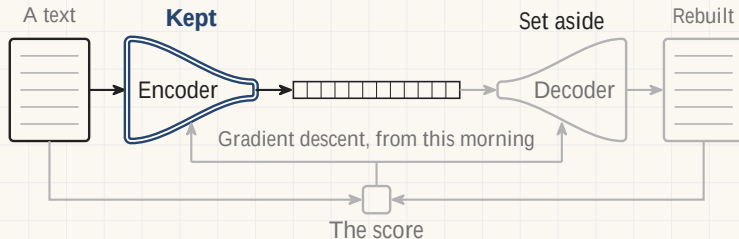
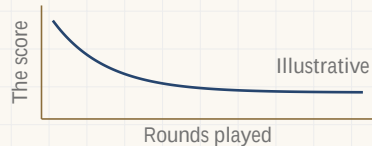
How it is trained

- 1 Collect a large collection of texts
- 2 Initialise the weights at random
- 3 Play the game, take the score
- 4 Update the weights from the score, play again

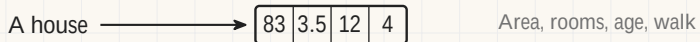


How it is trained

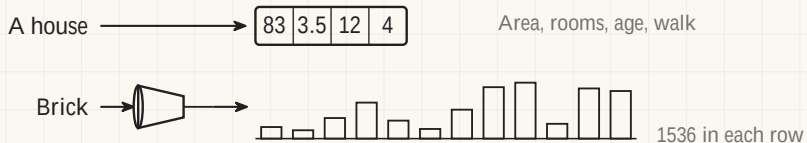
- 1 Collect a large collection of texts
- 2 Initialise the weights at random
- 3 Play the game, take the score
- 4 Update the weights from the score, play again



The numbers have no names



The numbers have no names



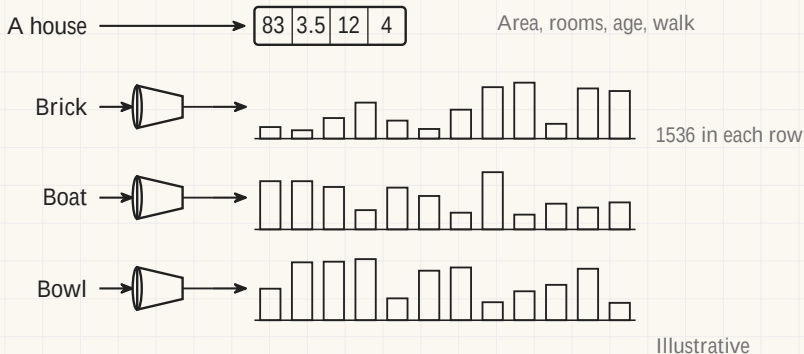
Illustrative

The numbers have no names

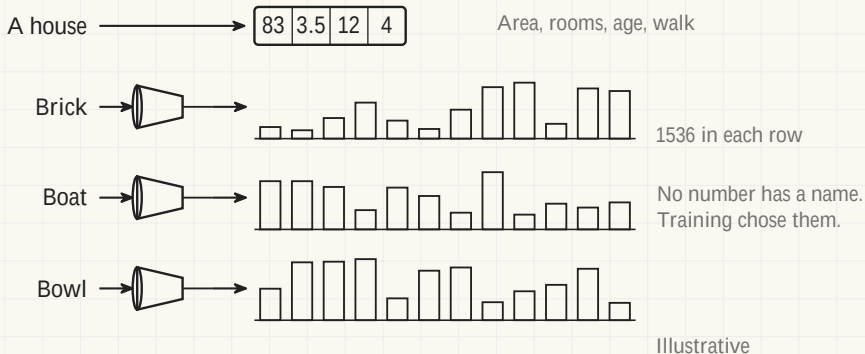


Illustrative

The numbers have no names



The numbers have no names

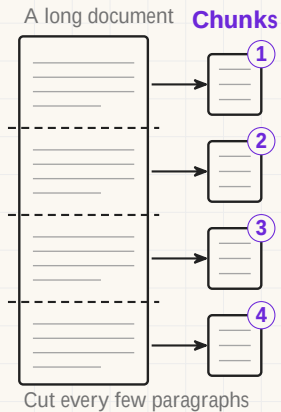


One key per chunk

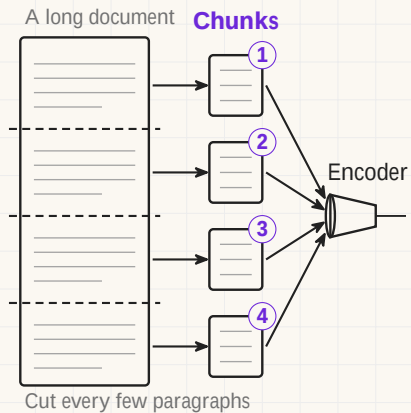
A long document



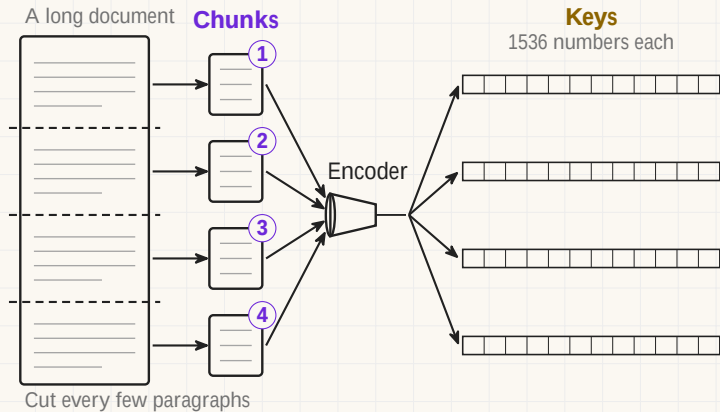
One key per chunk



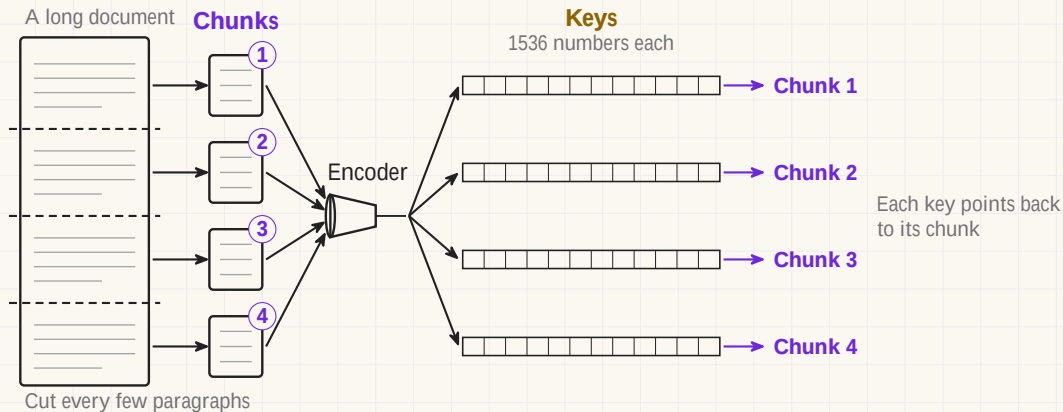
One key per chunk



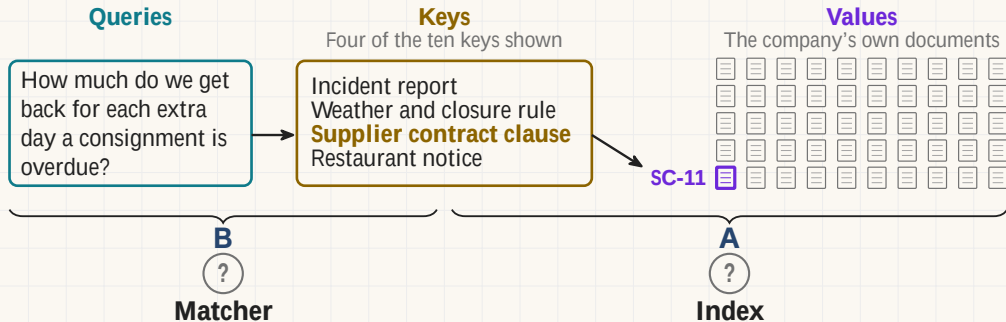
One key per chunk



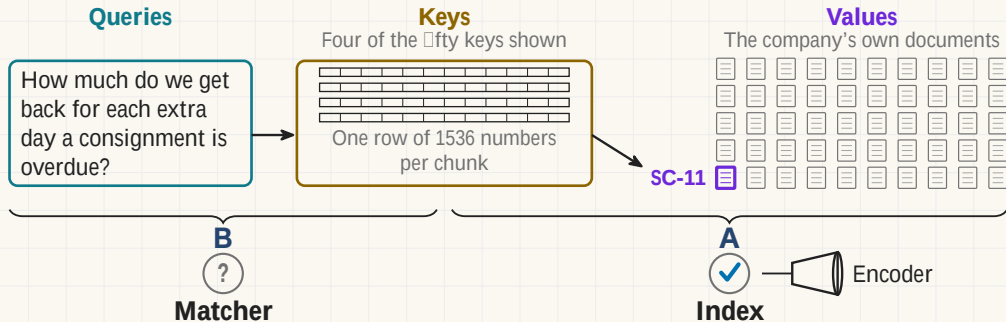
One key per chunk



Queries, keys and values



Queries, keys and values



6

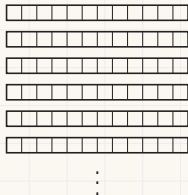
Section 6

Finding the relevant chunks

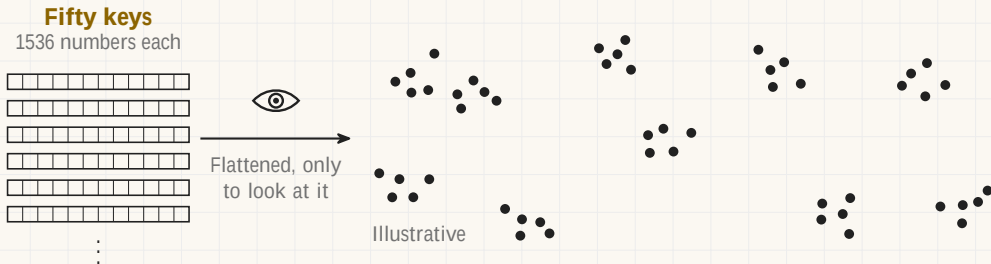
How do we effectively find the relevant keys?

Fifty keys

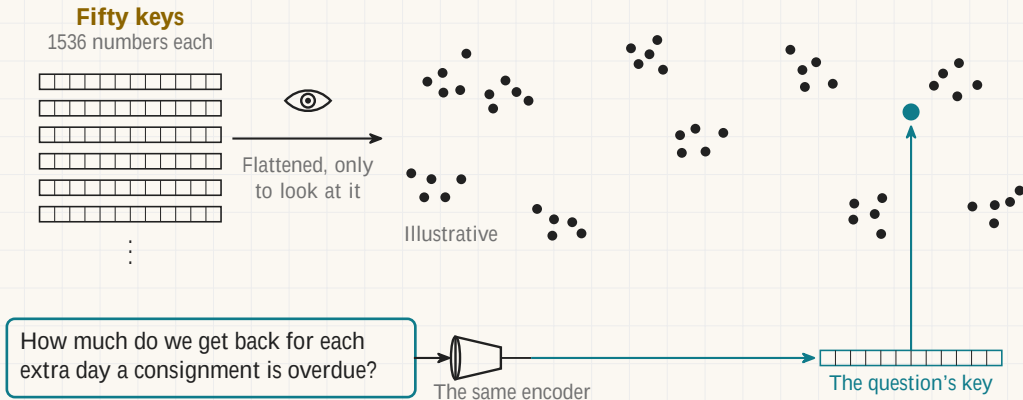
1536 numbers each



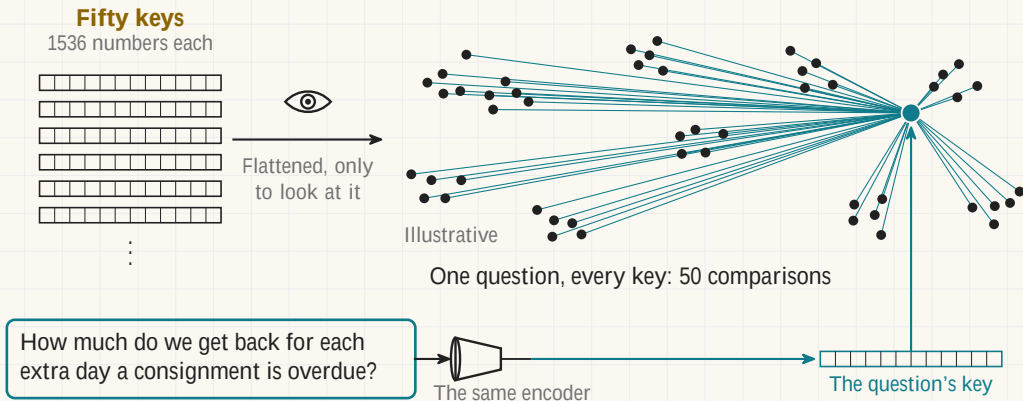
How do we effectively find the relevant keys?



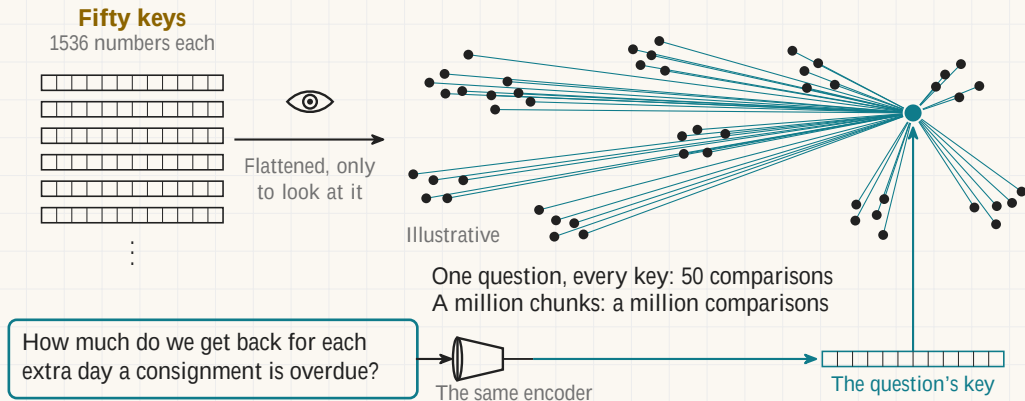
How do we effectively find the relevant keys?



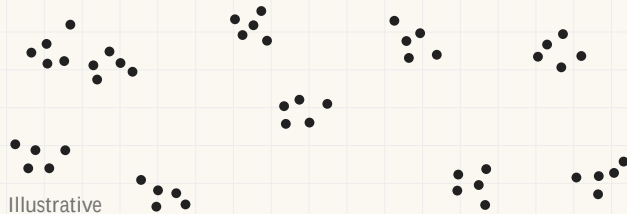
How do we effectively find the relevant keys?



How do we effectively find the relevant keys?

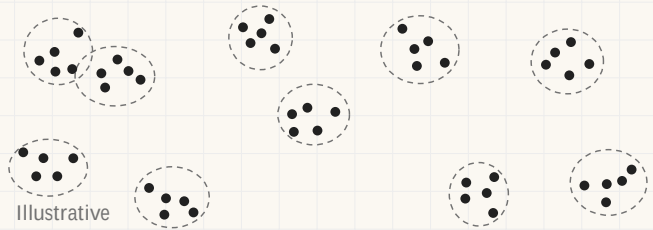


Done once, before any question



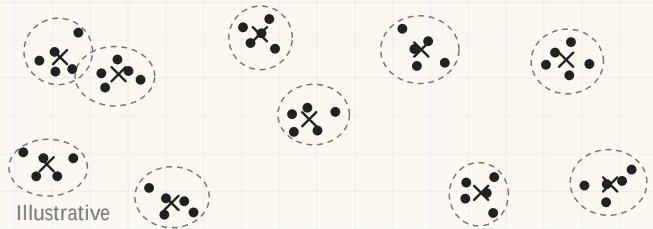
Done once, before any question

Yesterday's k-means



Done once, before any question

Yesterday's k-means
One centre per clump

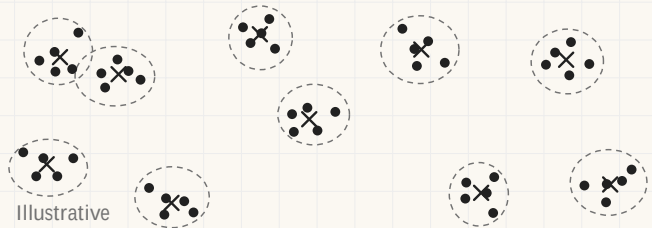


Done once, before any question

Yesterday's k-means
One centre per clump

×
×
×
×
×
×
×
×
×
×

← Kept



Done once, before any question

Yesterday's k-means

One centre per clump

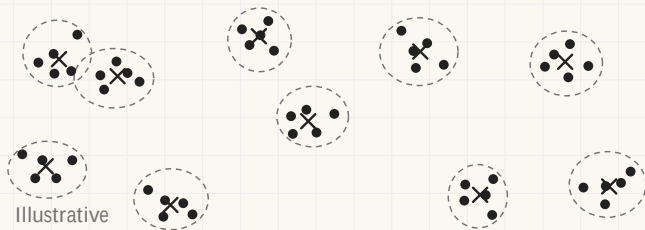


Kept

Illustrative

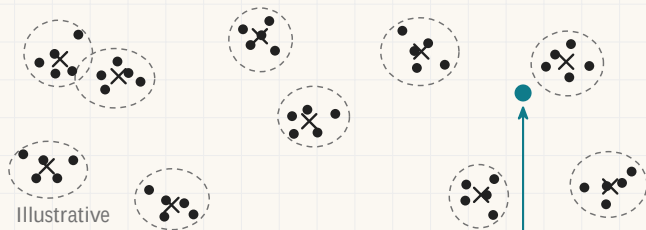
Each centre keeps the list of its chunks

At question time



How much do we get back for each extra day a consignment is overdue?

At question time



How much do we get back for each extra day a consignment is overdue?

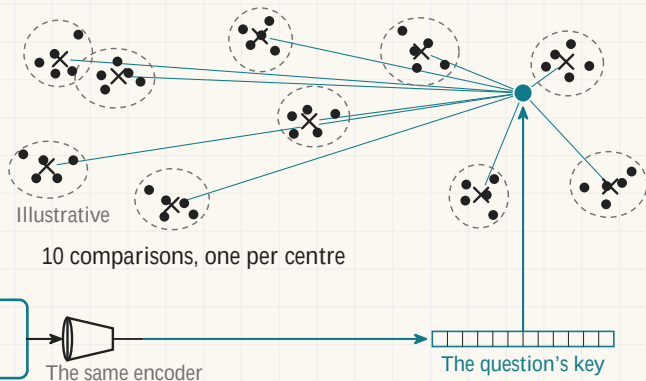


The same encoder

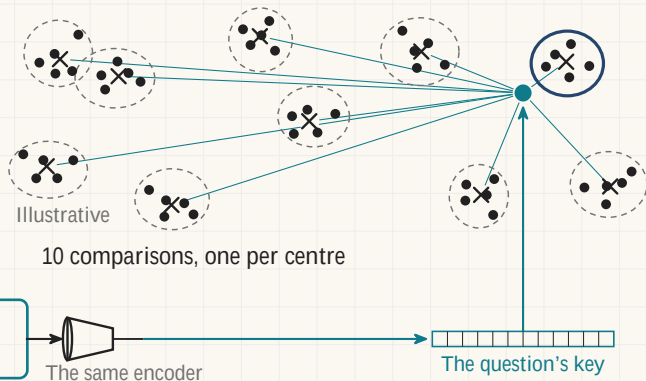


The question's key

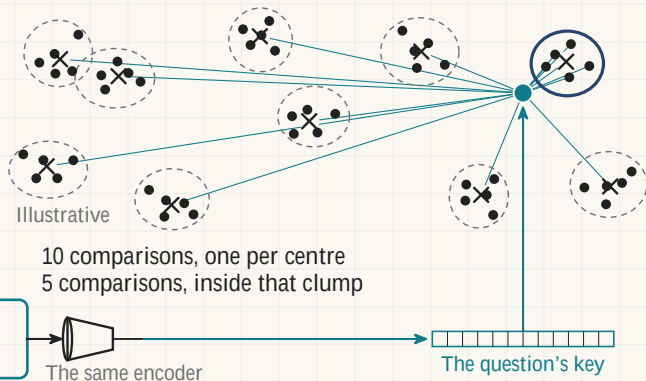
At question time



At question time



At question time



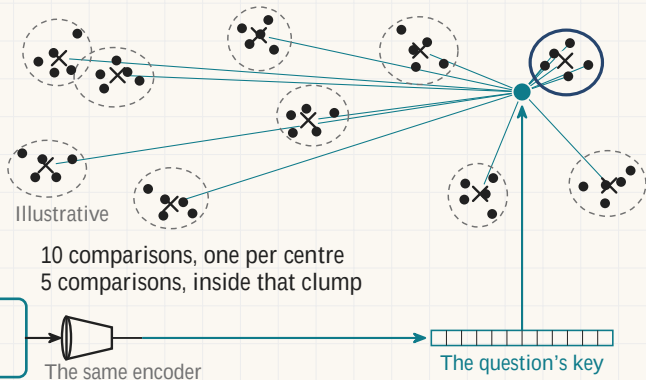
At question time

The chunks behind the closest keys



SC-11

15 comparisons instead of 50



Illustrative

10 comparisons, one per centre
5 comparisons, inside that clump

The same encoder

The question's key

At question time

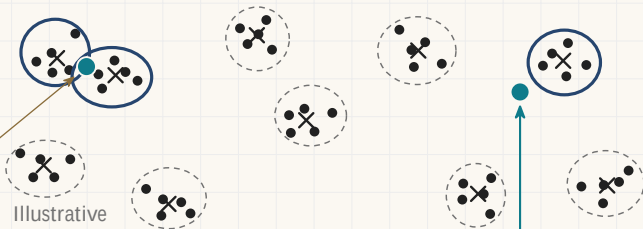
The chunks behind the closest keys



SC-11

15 comparisons instead of 50

A question near a border: its answer can sit in the next clump.
Real systems open more than one.



Illustrative

10 comparisons, one per centre
5 comparisons, inside that clump

How much do we get back for each
extra day a consignment is overdue?



The same encoder



The question's key

7

Section 7

Retrieval augmented generation

The full system

How much do we get back for each extra day a consignment is overdue?

The full system

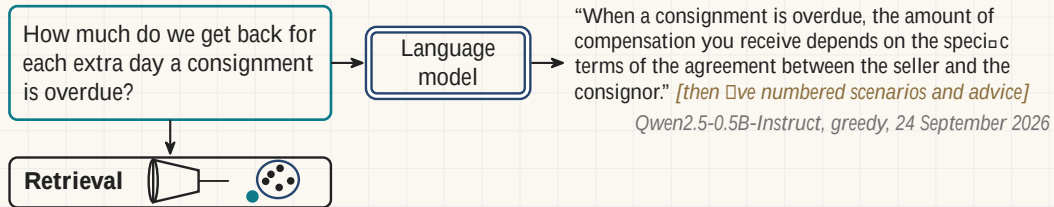
How much do we get back for each extra day a consignment is overdue?

Language model

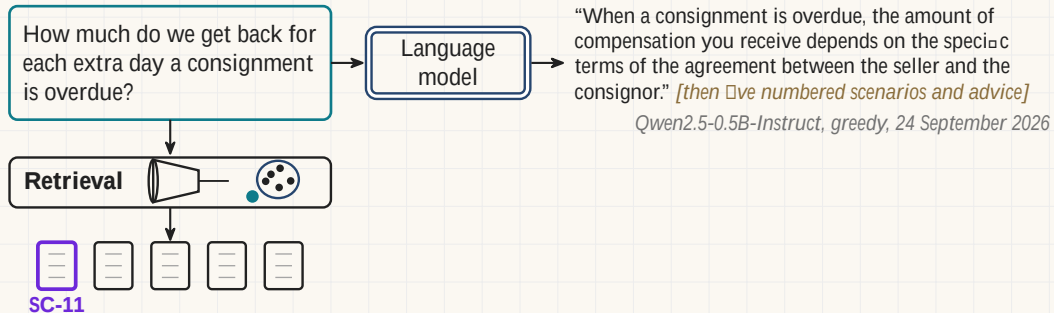
“When a consignment is overdue, the amount of compensation you receive depends on the specific terms of the agreement between the seller and the consignor.” *[then give numbered scenarios and advice]*

Qwen2.5-0.5B-Instruct, greedy, 24 September 2026

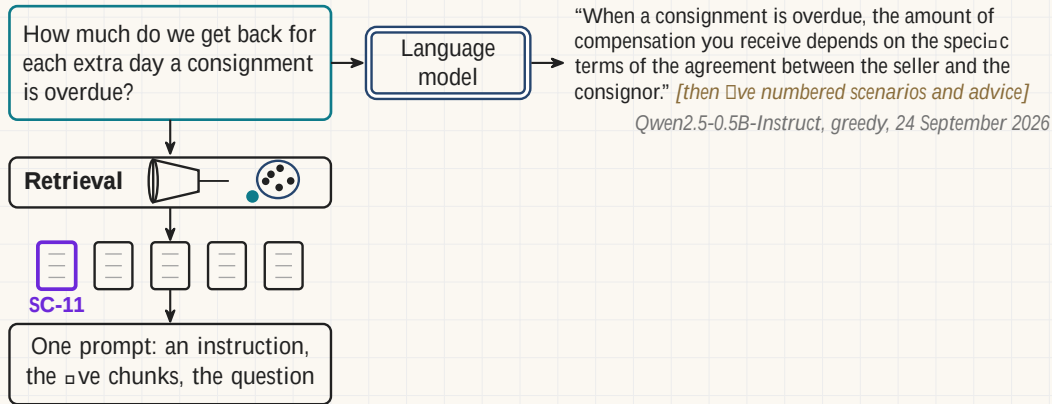
The full system



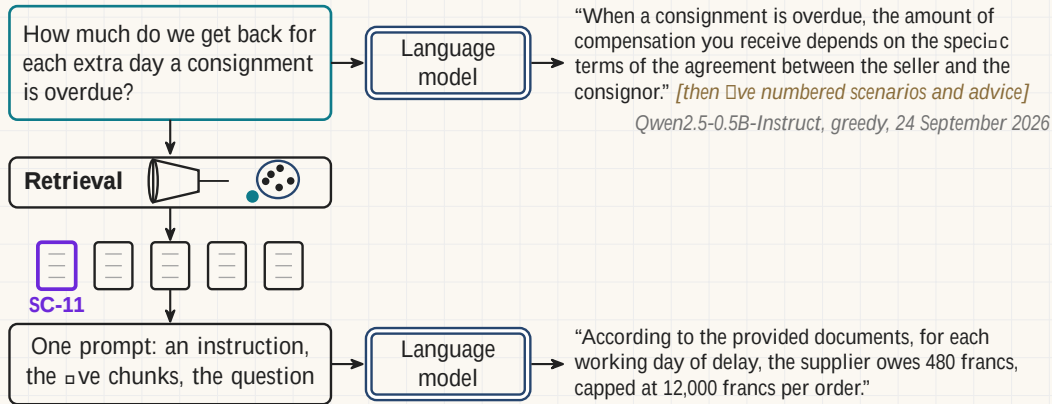
The full system



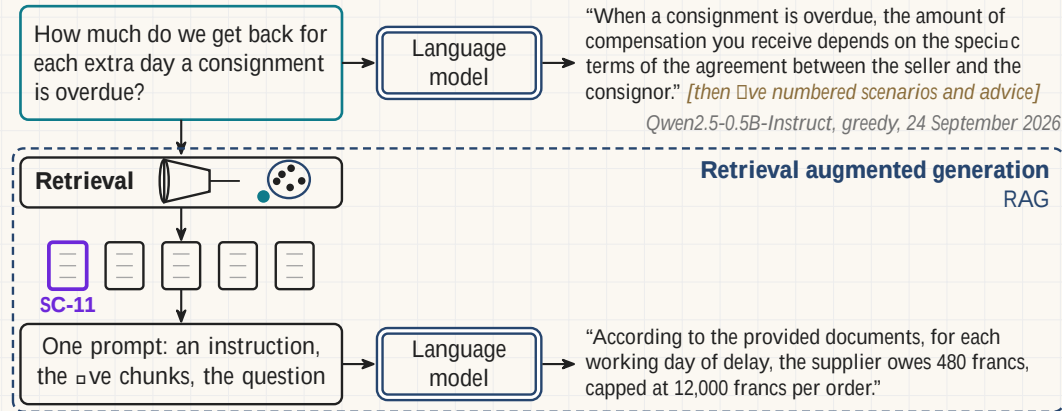
The full system



The full system



The full system



The prompt we send to the model

Written once, by a person

Answer the question using only the documents below. If the documents do not contain the answer, say so.

The prompt we send to the model

Written once, by a person

Answer the question using only the documents below. If the documents do not contain the answer, say so.

Found by retrieval

Document 1. Kernli Seiltechnik AG, framework agreement 2024.
Clause 6.2. Replacement haul ropes are delivered to Vorlen Dorf within 21 working days of a written order. For each working day of delay the supplier owes 480 francs, capped at 12,000 francs per order. Clause 6.3. The warranty on splicing work is 36 months from handover. Clause 6.4. Orders above 50,000 francs need a second signature.

[documents 2 to 5, four more chunks]

The prompt we send to the model

Written once, by a person

Answer the question using only the documents below. If the documents do not contain the answer, say so.

Found by retrieval

Document 1. Kernli Seiltechnik AG, framework agreement 2024.
Clause 6.2. Replacement haul ropes are delivered to Vorlen Dorf within 21 working days of a written order. For each working day of delay the supplier owes 480 francs, capped at 12,000 francs per order. Clause 6.3. The warranty on splicing work is 36 months from handover. Clause 6.4. Orders above 50,000 francs need a second signature.

[documents 2 to 5, four more chunks]

Asked by the user

Question: How much do we get back for each extra day a consignment is overdue?

The prompt we send to the model

Written once, by a person

Answer the question using only the documents below. If the documents do not contain the answer, say so.

Found by retrieval

Document 1. Kernli Seiltechnik AG, framework agreement 2024.
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[documents 2 to 5, four more chunks]

Asked by the user

Question: How much do we get back for each extra day a consignment is overdue?

What the model wrote

According to the provided documents, for each working day of delay, the supplier owes 480 francs, capped at 12,000 francs per order.

*Qwen2.5-0.5B-Instruct, greedy,
24 September 2026*

The prompt we send to the model

Written once, by a person

Answer the question using only the documents below. If the documents do not contain the answer, say so.

Found by retrieval

Document 1. Kernli Seiltechnik AG, framework agreement 2024.
Clause 6.2. Replacement haul ropes are delivered to Vorlen Dorf within 21 working days of a written order. For each working day of delay the supplier owes 480 francs, capped at 12,000 francs per order. Clause 6.3. The warranty on splicing work is 36 months from handover. Clause 6.4. Orders above 50,000 francs need a second signature.

[documents 2 to 5, four more chunks]

Asked by the user

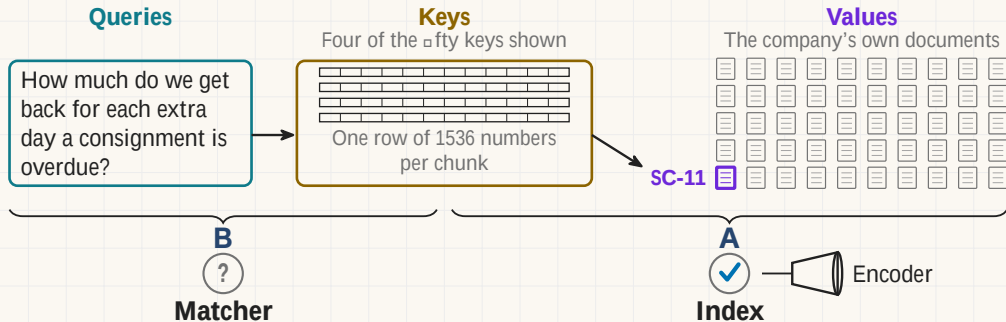
Question: How much do we get back for each extra day a consignment is overdue?

What the model wrote

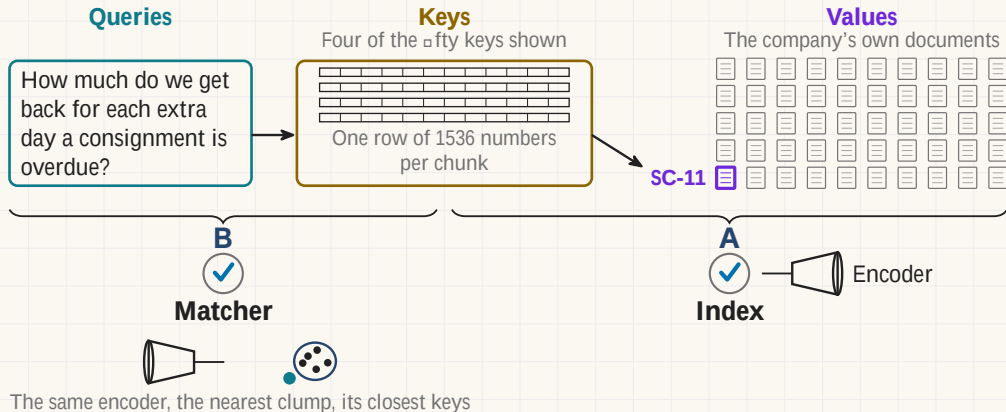
According to the provided documents, for each working day of delay, the supplier owes 480 francs, capped at 12,000 francs per order.

*Qwen2.5-0.5B-Instruct, greedy,
24 September 2026*

Queries, keys and values



Queries, keys and values



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